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To cite this article: Tathagata Banerjee, Diptarka Ghosh & Diptesh Das (2026) A state-of-the-art review and future challenges on application of devices for vibration control of bridges, *Mechanics of Advanced Materials and Structures*, 33:1, 2462713, DOI: [10.1080/15376494.2025.2462713](https://doi.org/10.1080/15376494.2025.2462713)

To link to this article: <https://doi.org/10.1080/15376494.2025.2462713>



Published online: 12 Feb 2025.



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ORIGINAL ARTICLE



A state-of-the-art review and future challenges on application of devices for vibration control of bridges

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ABSTRACT

Bridges are vulnerable to significant damage from both environmental factors and human-induced threats. Recently, vibration control has emerged as a crucial method for minimizing bridge damage. The present study conducts a methodical examination of literature on diverse vibration control devices for mitigating of bridge responses. The study categorizes the control devices and extensively compares various control devices, emphasizing their respective strengths and limitations. A prevailing shift is being observed toward semi-active and hybrid control devices from the traditional passive or active control devices. Based on the present review of literature, recommendations have made for as future scope of research.

ARTICLE HISTORY

Received 20 June 2024
Accepted 31 January 2025

KEYWORDS

Bridges; vibration control; passive control; active control; semi-active control; hybrid control

1. Introduction

Bridges play a major role in the smooth operation of our daily lives. Different bridge designs and layouts offer unique benefits and drawbacks as well as specific uses. The crucial challenge of bridges is to support properly against dynamic loads, such as those of cars, trains, blasts, or multi-hazard events. Multi-hazard events, such as cyclones, earthquakes, or blasts, can severely damage the overall structure due to the extreme vibration. In addition, as a result of fatigue and creep, bridges typically lose a significant portion of their load-bearing capacity throughout the course of their service lives, making the structure highly susceptible to failure. One of the major challenges facing researchers is mitigating such large vibrations that are induced by different multi-hazard events. Numerous analytical and laboratory-based studies have been conducted over time to strengthen bridges against random excitations. Dampers, isolators, and sensors are utilized in an individual or combined manner to improve this strengthening procedure.

The damage loads on the bridge structures are irregular due to a variety of natural hazards. One of the recommended methods for reducing the unintended actions due to these irregular loads is to alter the bridges dynamic properties with regard to a certain load. Yao [1] introduced the concept of structural control for the first time in 1972. Bridge vibrations are caused by a variety of complicated hazards, including wind and seismic excitations. To mitigate these vibrations, vibration control devices are employed for structures. Over the last few decades, vibration control of bridge has been a popular, considerable, and expanding area of research. There are two methods for providing structural control: (I) employing smart materials in bridge [2]; and (II)

installing control devices on bridges, such as dampers, and isolators [3–5]. Numerous publications on the issue of mitigating vibration in bridge constructions, which are susceptible to dynamic loads have recently been published. The purpose of this study is to show the improvement over the decades of the performances of the structure against dynamic loading through investigating the types and applications of different devices in bridges. This paper reviews the use of several isolator and damper in bridge vibration control. Many factors, such as excessive vibration due to hazards (e.g. earthquake, wind), design of control system (e.g. kinds, numbers, and placements of control devices), etc., influence a bridge vibration control system's efficiency [6]. Housner et al. [2] provided brief research in 1997 that covered the potential of control theory in structural vibration control as well as passive, semi-active, active, and hybrid control systems.

Fisco and Adeli [7] presented an overview of structures with active control algorithms, including semi-active stiffness dampers, distributed actuators, active magnetorheological (MR) dampers, active tuned mass dampers (TMD), active tendon systems, piezoelectric sensors, etc. Fisco and Adeli [8] conducted a review on hybrid control systems combining semi-active device with base isolators, hybrid mass damper, and semi-active tuned liquid column dampers (TLCD) with passive dampers, with a focus on the application of useful and dependable control algorithms, such as the linear-quadratic-gaussian technique, linear-quadratic-regulator [9–12] and sliding mode [13]. Active vibration control systems in structures, such as active cable, aerodynamic appendages, active tuning mass, and tendon control, and active structural control, were studied by Korkmaz [14] using a range of

control algorithms applied to actuators and sensors. An overview on active TLCD, active TMD, and piezoelectric actuators for controlling vibration of structures under dynamic loads was published by El-khoury and Adeli. In addition, TLCD, circular TLCD, tuned liquid column ball dampers, and pendulum TLCD were studied in the context of passive vibration control. The semi-active segment was also described with focus on the negative stiffness devices, semi-active TMD, decentering variable friction damper, and MR dampers. In the category of hybrid systems, semi-active magnetorheological damper with non-linear base isolator and active base isolation system were also included [15]. In their overview of tuned mass dampers (TMD), Gutierrez Soto and Adeli [16] categorized the main types as passive TMDs, passive tuned liquid column dampers, bidirectional TMDs, and pendulum TMDs to investigate the potential locations for control devices in structures using a range of dynamic excitations. Basu et al. [17] explored the utilization of control devices for vibration control of bridges, wind turbines, and other types of structures in Europe, considering the effectiveness of functional materials as well as passive, active, and semi-active dampers.

Amezquita-Sanchez et al. [18] conducted a review of smart material-based control systems for structural vibration mitigation, focusing on the benefits and drawbacks of smart systems as well as control strategy trends in smart civil structures. Lu et al. [19] investigated advanced technologies for nonlinear dissipative devices, including energy sinks, viscous dampers, and particle impact dampers to control structural vibrations. Ma et al. [20] reviewed the uses of passive inerter-based devices for structures and compared their benefits to traditional vibration control methods. Tabrizikahou et al. [21] discussed SMA-based isolation devices and their limitations in vibration control of structures. Javanmardi et al. [22] provided a comprehensive analysis of structural vibration control systems for cable-stayed bridges. As an outcome of their review, key characteristics of each control approach were presented and several types of devices and their designs were explored to identify the optimal control method for reducing cable vibrations. Sun et al. [23] provided a thorough assessment of recent improvements in stay cable vibration reduction, including mathematical modeling of the cable damping system and approaches for improving multimode damping.

Since control of bridge vibrations is the need of the hour and involves different types of complexities, it has drawn the attention of researchers and engineers world over. This has led to considerable number of theoretical and experimental studies, with the emergence of many new and innovative vibration mitigation techniques. The control device is the most important component through which the control scheme is applied on a structure to obtain the desired reduction of its responses. These devices ensure that processes or operations run efficiently, safely, and within the intended parameters. However, there is a dearth of review papers on the devices used in the different control schemes. Even though there are few review papers on vibration control devices, these studies have concentrated on particular

types of control device for different types of structures. There has never been a comprehensive review paper on all the existing control devices for bridges, which is one of the primary lifeline infrastructures. Therefore, in order to address the growing challenges involved, the present paper provides a comprehensive state-of-the-art review of the vibration control devices with the help of which the different control techniques are implemented. For the convenience of the readers, an overview of the overall concept following the standard classification of passive, semi-active, active, hybrid vibration control devices for various types of bridges is presented. The developments from initial implementations to recent upgradations of structural control devices used in bridges are provided under each category. The highlights of each research, namely, the innovative contribution, the advantages and drawbacks are discussed in detail. For each type of device, the real-life applications, stating the key features, are also provided. In the present study, various types of bridges like beam bridges, suspension bridges, truss bridges, foot bridges, and cable-stayed bridges subjected to multi-hazard loads are taken into account. After a thorough discussion as mentioned above, the paper finally summarizes the review and based on the gap areas identified, proposes future research scope in the given area. The recommendations are expected to be helpful for the researchers working in the field of vibration control of bridges.

The current article has been organized in this manner. In [Section 2](#) passive, semi-active, active and the devices for hybrid vibration control systems are thoroughly explained based on their operating mechanism and benchmark real-life applications are presented. A comparison of bridge vibration control devices is summed up in [Section 3](#), and [Section 4](#) concludes with relevant recommendations for future research endeavors.

2. Structural control devices for bridges

A structural control system is a device that reduces undesired vibrations to improve the performance and life-span of physical components. Over the years, several structural control analytical and experimental investigations have been carried out to find out a reliable structural control system. Accurate modeling the bridges is crucial for the analytical investigation of the bridge structure. Primarily, lumped-mass modeling [24, 25] and finite element modeling [26, 27] are two methods for bridge modeling. As an alternative to finite element method (FEM) analysis, Carrera proposed Carrera Unified Formulation (CUF) based FEM analysis [28–36]. CUF based FEM analysis has recently attracted the attention of researchers because it reduces time consumption and increases the effectiveness of numerical analysis [37–53]. For bridge analysis, CUF based FEM analysis might turn out to be a reliable option in the future, aiding in the exploration of more dependable bridge vibration control systems. There are two approaches of vibration control strategies: either connecting vibration control devices to the structure or directly incorporating the vibration control system within the structure, which can be done with

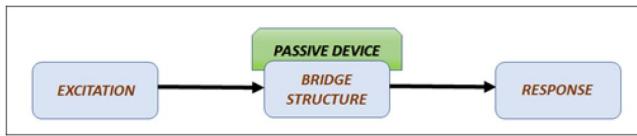


Figure 1. Simplified representation of bridge vibration control using passive devices.

metamaterials. Many researchers have investigated on innovative mechanical metamaterials [54–61]. The concept of mechanical metamaterials and their applications for vibration control was discussed by Dalela et al. [62]. There are four main categories of structural control systems: passive, semi-active, active and hybrid vibration control.

2.1. Devices for passive vibration control system

Passive vibration control systems represent the most fundamental type of structural control systems that operates without relying on any external power supply. These systems, by nature, do not inject energy into the structure, thereby providing stability purely by dissipation. Figure 1 illustrates a schematic diagram depicting a passive vibration control system applied to a bridge structure.

2.1.1. Bridge vibration control strategies using passive devices

Passive energy-dissipating systems employ mechanisms that attenuate energy from structural vibrations without relying on any external power source. They function as supplemental damping components, external to the structure. These passive systems generate control forces in response to the structure's motion intensity. Notably, no external energy is supplied, ensuring stability of the bridge. These control devices often boast simple designs, cost-effectiveness, and easy deployment.

Numerous passive control methods are employed to safeguard civil engineering structures against random vibrations such as earthquakes, cyclones, and man-induced disruptions like terrorist bomb blasts [63–67]. Significant progress in the conceptualization, design, specifications, and incorporation techniques has facilitated the widespread application of these devices in bridge engineering. Based on their ability to dissipate energy, passive control devices can broadly be segregated into three categories: seismic isolation (SI), passive tuned dampers, and passive viscous dampers.

2.1.1.1. Bridge seismic isolation (SI) systems. The SI method stands out as a commonly utilized passive control technique aimed at fortifying bridges against seismic forces. The bridge deck and pier are separated by carefully designed isolation systems. These devices introduce flexibility to the bridge deck at its base in the lateral direction, effectively mitigating the seismic impact on the superstructure by extending the bridge's time period. This characteristic renders it a potent tool for controlling vibrations in bridges as it enables the reduction of high natural frequencies with the aid of such devices.

Furthermore, owing to its nonlinear behavior and the development of a hysteretic loop, these isolators dissipate a substantial amount of induced energy by themselves. Numerous numerical studies in the realm of seismic isolation systems have affirmed their viability in safeguarding bridges against the motions generated by earthquakes.

Various isolators are employed in the vibration management of bridge structures, including laminated rubber bearings, lead rubber bearings (LRB), friction pendulum system (FPS), high-damping rubber bearings (HDRBs), resilient-friction base isolators, cylindrical rods, elliptical rolling rods, and shape memory alloys.

The passive control approach, which operates based on principles like period shift and increased energy dissipation, finds significant application in base isolation techniques. This technique leads to increased structural flexibility, reduced attraction of seismic forces, and enhanced structural damping. It is possible to increase the superstructure's natural period by applying a rubber like material having minimal lateral stiffness, so as to enhance the flexibility of the structure. Mostaghel and Khodaverdian [68] suggested the use of a resilient-friction base isolation (R-FBI) system for bridge vibration. The center rubber core of this base isolator is concentrically surrounded by Teflon-coated layers. It incorporates the durability of rubber with benefits of friction damper. The rubber core efficiently distributes sliding velocity and displacement, along the height of R-FBI bearing. These components are bonded to the sliding ring and a load-bearing structure that isn't vertical. By integrating friction, damping forces, and spring characteristics, this system achieves effective isolation. Another seismic isolation device of conceptual interest is the FPS, constructed by combining sliding bearings with principles of a pendulum-type response. This system can be conceptualized for achieving maximum sliding friction coefficient by the use of an articulated slider on concave chrome surface and a spherical slider lined with a bearing material connected to a polished chrome surface [69].

The "Electric de France" (EDF) represents a notable category within friction-type seismic isolator systems. Such devices are designed for application in high seismic zones, particularly in nuclear power plants [70]. Lin and Hone [71] introduced a unique arrangement with free circular rolling rods positioned between the base and foundation. Advantageous trait of the bearing lies in its low rolling friction coefficient, which significantly reduces the transmission of earthquake forces to the superstructure. Nonetheless, because of the system's recentering capability, considerable peaks and residual deformations were observed. In addressing this issue, Jangid and Londhe [72] proposed the use of elliptical rolling rods instead of circular ones. This approach aimed to amalgamate the advantageous characteristics of both the EDF and R-FBI systems. They suggested replacing the elastomeric bearings used in the EDF's isolation with R-FBI units. In the modified design, the friction plate assumed the position of the upper surface of the R-FBI system. Consequently, bridge foundation became prone to movement, resembling the behavior of the EDF seismic isolation

system. Furthermore, FPS was equipped as a bearing within pier and deck of a multi-span continuous (MSC) bridge by Wang et al. [73]. It was concluded by their investigation that the excessive superstructure displacement caused by the sliding isolation, could completely be avoided by replacing one of the rollers supports in the abutments with a hinge support. Li [74] investigated the behavior of a standard three-span bridge equipped with isolation system comprising rubber bearings and longitudinal hysteretic dissipaters. To filter out the white noise ground acceleration, the stochastic response of the structure was computed using a comparable linearization technique based on nonlinear equations of motion obtained for primary mode. Additionally, a technique was proposed for designing an optimal vibration control device for bridges utilizing hysteretic dampers. Constantinou et al. [75] recommended a multidirectional sliding Teflon bearing system which ensured controlled displacement during earthquakes, and possessed the capability to recenter. These devices were found to be able to significantly dissipate energy while offering resistance to a rigid bridge deck against service loads. An assessment was conducted utilizing LRB to determine the equivalent damping ratio and to find out its effectiveness in bridge structures [76, 77]. Another model was employed to elucidate the concept of hysteretic restoring force in isolation bearings. Nonlinear responses to earthquakes for bridge structures was presented by combining Wilson-theta techniques with fourth order Runge-Kutta methods. As an example, an MSC bridge equipped with isolators was utilized to investigate the implications of isolation bearings [78]. Grassier and Cozzarelli [79] examined Nitinol shape memory alloys (SMAs) as seismic dampers, studying how loading history and frequency influenced the dissipation of energy in Nitinol wires. For pseudo-elastic behavior, they proposed an experimentally tested one-dimensional constitutive model. An analytical investigation was carried out to study a SI system incorporating SMA device. This analysis evaluated the capability of the proposed model to minimize displacement and dissipate seismic energy effectively [80]. Additionally, Adachi and Unjoh [81] conducted a micro-model analysis on SMA plate damper, employed for bridge energy dissipation. Their research highlighted the significance of the martensitic phase in shape memory alloys, contrasting it with the austenitic phase, showcasing the most advantageous application. Multiple researchers have proposed the use of SMAs in specific bridge parts to enhance post-earthquakes longevity. Some proposed structural solutions include bridge girders or decks with prestressed SMA wires [82–84], rocking bridge piers [85], and SMA-based bridge columns [86–90]. SMA has found application in specialized devices aimed at enhancing structural performance. In bridge columns, SMA rebars were utilized for active confinement and prestressing, while SMA wires were employed for longitudinal reinforcing and controlled yielding. Furthermore, research has been focused toward exploring potential SMA applications in isolation bearings and retainers [91]. Numerous studies have investigated the use of SMAs as damping devices in stay cables, expansion joints, and

post-tensioning tendons in girders. These studies highlighted the frequent utilization of isolation methods to mitigate the structural responses. The efficiency of various seismic isolators was assessed for a MSC bridge. A parametric study was conducted to determine the optimal design parameters for different isolators, including laminated rubber bearings, N-Z bearings, pure friction, resilient-friction seismic isolators, and EDF devices. The primary subjects of their investigation were the primary time-period of the bridge and bearing's friction coefficient. In the comprehensive study of various isolators, resilient friction isolators emerged as the least sensitive to fluctuations in seismic intensities and frequencies [92]. Jangid [93] specifically explored the response reduction capabilities of lead rubber bearings in a bridge under near-field earthquakes. A bilinear model was presented in their study to represent force-deformation of LRB. The changes were observed in the structural responses with variations in the flexibility isolation period of the superstructure and the yield strength of the bearings. It was indicated in their study that an optimal value for the yield strength of the bearing is minimized when dealing with near-field earthquakes. Subsequently, a parametric investigation was conducted by Hameed et al. [94] on the response of isolated bridges under earthquakes, differing the parameters of LRB. The variables taken into consideration in their study included the yield strength relative to the total weight acting on the bearing, stiffness of the bearing, and the ratio of yield stiffness to post-yield stiffness. The research concluded that there exists a specific range of values within which the structure's response remains minimized. A sensitivity analysis of the mechanical characteristics of lead core rubber bearings was performed by Ahmadipour and Alam [95]. As a result of extensive research, recent advancements for achieving higher accuracy, efficiency, and adaptability to complex and dynamic environments has led to new and innovative isolation techniques [96–99].

Several investigations [100–107] have further supported the benefits of SMA's super-elastic behavior. While the majority focus on Ni-Ti SMA, Hosseini et al. [108] specifically explored the effects of utilizing Cu-based SMA to reinforce plastic hinges in bridge piers. They employed Cu-Al-Mn SMA to analyze bridge piers subjected to combine axial, flexural, and shear stress. Despite observing minimal residual deformation, they noted an inadequate energy dissipation capacity. Bhuiyan and Alam [109] compared the seismic performance of a three-span highway bridge during mild to severe seismic events. It was observed that, when piers were subjected to mild seismic events, the installation of SMA bearings significantly decreased residual deformation. However, the performance significantly declined during severe seismic events. To address the limitations of earlier systems, Dezfuli and Alam [110] introduced a unique arrangement of isolators utilizing diagonally oriented SMA wires and rubber-based bearings. After that, numerous investigations was conducted with various designs of isolation bearings based on SMA wires. These studies collectively concluded that SMA-based isolation devices exhibit superior recentering abilities and energy dissipation capabilities

[111–117]. Masuda et al. [118] utilized FEM to develop a constitutive model to investigate the suitability of SMA springs for use in isolation devices. SMA bars were incorporated by Wilde et al. [80] with laminated rubber bearings, who then evaluated the performance of the bearings against lead rubber bearings. Their study revealed that the combination of rubber bearings and SMA bars enhanced energy dissipation while reducing susceptibility to bridge vibrations. Recent research efforts aiming for improved vibration reduction devices have focused on combining multiple passive vibration control devices. For instance, some studies have explored the integration of SMA-UDs (Shape Memory Alloy U-shaped Dampers) within LRBs featuring a lead core and steel U-shaped damper [119], as well as SMA-LRB systems [117, 120], and SMA-HDRB (High-Damping Rubber Bearings) configurations [121].

2.1.1.2. Passive tuned damper. A single tuned-mass damper (TMD) was proposed by Frahm [122]. The amplitude of the primary system for a single frequency could almost be reduced entirely by a mass-spring absorber, which is used to control the rolling motion of ships. Das and Dey [123] examined the extent to which TMDs decreased the random response of simple bridges using the finite element method. They argued that the Eigen frequency distribution of the structure largely determines the efficiency of a particular TMD design. Lin et al. [124] examined the optimization of TMD parameters aimed at diminishing wind-induced responses in long-span bridges. Researchers explored the utilization of Multiple Tuned Mass Dampers (MTMDs) to manage vibrations caused by wind in such bridges. Gu, Chen, and Chang [125] performed a parametric analysis of MTMDs to control buffeting in the Yangpu Bridge. Pourzeynali and Datta [126] demonstrated the efficacy of TMDs in mitigating flutter in suspension bridge decks, asserting that the maximum critical wind speed causing significant flutter in suspension bridges is determined by factors such as damping ratio, tuned frequencies, and the number of TMDs. Wang, Lin, and Chen [127] showcased how TMDs effectively control vibrations, minimizing absolute accelerations, end rotations, and vertical displacements induced by train accelerations at resonance speeds on bridges. Meanwhile, Chen and Cai [128] presented an innovative control strategy to alleviate the impact of modal coupling in addition to reducing resonant vibrations using TMDs. Their findings revealed that in cases where coupling effects were substantial and damping ratios in the specific concerned modes were significant, the newly suggested controller outperformed the conventional resonant suppression approach in terms of controlling vibrations more effectively. Lin et al. [129] highlighted the superior performance of MTMDs over single TMDs. Mottershead and Ram [130] synthesized various inverse techniques utilized in passive alteration and control, offering direct formulations to substitute observed responses with unit-rank adjustments. Chen and Wu [131] specifically delved into assessing MTMDs' efficacy in damping vibrations induced by wind in bridge structures. Hoang et al. [132] proposed a deck-isolator

combination to function as a TMD, emphasizing the success of a high mass ratio TMD in reducing a bridge's reactions while maintaining resistance to ambiguity in system parameters. Luu et al. [133] suggested that MTMDs effectively mitigate vibrations in bridges caused by high-speed trains. Casalotti et al. [134] introduced hysteretic TMD systems aiming to mitigate multi-mode flutter in long-span suspension bridges. Matin, Elias, and Matsagar [135] presented the utilization of MTMDs to control continuous concrete bridges across multiple modes. Additionally, Pisal and Jangid [136] concluded that optimized multiple tuned mass friction dampers positioned at the mid-span of a bridge outperform both optimized single tuned mass friction dampers and multiple dampers spread along the bridge length in reducing response. Tubino et al. [137] installed two TMDs to regulate vibration on a footbridge under varying loading events. It was shown in their investigation that under almost all loading situations, the installation of two TMDs greatly reduced the vibration's amplitude, leading to the attainment of adequate satisfaction levels. A single site-pounding tuned mass damper (SS-PTMD) was used in an experimental investigation to mitigate the vibrations induced by vortex on a bridge deck. Their experimental findings displayed that the application of an SS-PTMD with a mass ratio of 2% lowered the maximum response of the bridge deck model by 94%. Further, it was demonstrated that nonlinearity in vortex-induced forces does not significantly affect the control efficacy of optimized SS-PTMD under sinusoidal excitation [138]. The nodal mode vibration of the cable using the SS-PTMD under several loading scenarios, such as harmonic and generated turbulent wind excitations, was investigated by Wang et al. [139]. Additionally, field testing were conducted on the A10 cable of the Dongting Lake Bridge in order to verify the SS-PTMD's performances.

In order to manage the structural displacement of a three-span continuous bridge against earthquakes, Elias et al. [140] presented a study to set up the parameters of the non-linear tuned vibration absorbers (n-TVAs). Jiang et al. [141] scrutinized the seismic responses of a single span bridge structure equipped with KDampers subjected to spatially varying ground motions (SVGMs). Negative stiffness element K_{Negative} was established by the K_{Damper} between the ground and the auxiliary mass of TMD and positive stiffness element K_{Positive} between the superstructure and auxiliary mass. According to the analytical findings, the optimized KDampers were successful in lowering the bridge's displacement and absolute acceleration responses. In order to mitigate the vortex induced vibration (VIV) in long-span bridges, Xu et al. [142] presented four distinct inerter-based dampers (IBDs), each of which had a particular mix of dashpot-spring components in series or in parallel with the inerter device. Their study illustrates the advantages of IBDs over traditional TMDs in bridge VIV control and suggests two feasible IBD configurations for bridge VIV mitigation. An inertial device and conventional tuned mass dampers (TMD) are combined in a device called a tuned mass damper inerter (TMDI). Instead of using identical TMDIs, Dai et al. [143] applied multiple tuned mass

damper inerter (MTMDIs) to provide an adequate equivalent damping ratio against the unpredictable vortex-induced vibrations (VIV) in flexible bridges across a wide lock-in wind speed range. Over a wide lock-in wind speed range, the well-optimized MTMDI produced a more reliable control result than the TMDI.

Recent developments in refining TMDs [144] and other inerter devices [145] for better accuracies are in progress with multiple novel and advanced techniques being suggested, that are yet to be implemented for bridge applications.

2.1.1.3. Passive viscous dampers. Viscous dampers operate as velocity-restraining devices by generating force through the flow of high-viscosity fluid *via* specific orifices. Fluid viscous dampers (FVDs) are supplementary energy dissipation devices that have long been used to protect bridges from earthquakes [19]. An efficient FVD requires optimal design, placement, and sizing. In recent decades, the utilization of FVDs has become increasingly prevalent in passive control systems for bridges. This is owing to their capacity to notably enhance seismic performance by dissipating energy, producing forces out of phase with displacements, and elevating the damping ratio of the structure without notably altering its stiffness [146, 147]. This characteristic spares the typical successive hit-and-trial process often associated with other passive control devices such as hysteretic dampers or seismic isolators [148–151]. Viscous dampers can be broadly categorized into two types based on the methods employed to generate damping force: rod-type viscous dampers and viscous damping walls [147, 151].

In scenarios where long-period structures experience near-fault earthquakes, the linear viscous damper (LVD) may generate excessive damping force directly correlated with the relative velocity, potentially rendering the device inefficient. This inefficiency might stem from the insufficient strength of connectors, leading to failure in fulfilling its intended purpose [152]. Conversely, nonlinear viscous dampers (NVDs) tend to produce damping force that increases marginally at low or moderate velocities (as seen in frequent earthquakes) and does not significantly escalate with higher velocities (common in rare earthquakes). This difference shows that NVDs can reduce damping force far more than LVDs can while keeping the damping effect, which makes their design and manufacture more reasonable and affordable [153, 154]. Furthermore, NVDs exhibit the capability to provide sufficient damping efficacy for various load durations, such as short-duration shock loads and long-duration wind and earthquake loads, thus showcasing their versatility [155]. NVDs are effective in mitigating the impact of higher modes on structural reactions [156, 157]. According to data from the Taylor Company on structural applications utilizing fluid viscous dampers, as of 2015, 592 engineering projects worldwide had incorporated these dampers, with 24.2% of them used in bridge applications [19].

Various classical laminate theories, including first-order and higher-order shear deformation theories as well as

zig-zag theories, were assessed for modeling a layered composite structure with a viscous core. Comprehensive static and dynamic analyses within the linear range showed that models based on zig-zag theories provided greater accuracy, especially in cases where the stiffness ratio between the core and the faces was significant. Carrera [158] provided a detailed overview of zig-zag theories employed in the analysis of laminate structures, including plates and shells. To establish a comprehensive framework capable of deriving nearly all structural theories as specific instances of a generalized arbitrary expansion of unknowns, the Carrera Unified Formulation (CUF) was introduced [159–163]. This unified approach accommodates a wide spectrum of material behaviors and geometric configurations, including nonlinear effects and complex structural interactions. CUF offers significant computational efficiency by reducing computational costs without compromising accuracy. Its versatility was demonstrated in various applications, such as analyzing beams with layers or inclusions of viscoelastic materials [164]. Furthermore, CUF-based one-dimensional theories have been employed to evaluate viscoelastic structures with delamination [165], focusing on the modal damping characteristics influenced by delamination effects. Soltane et al. [67] introduced an adaptive nonlinear model for a viscous damper that mitigated vibrations of a stay cable of the Rades-La Goulette cable-stayed bridge in both forced and free vibration situations. They demonstrated the superiority of the introduced model over the basic Maxwell model.

Zhong et al. [166] presented an optimization methodology that uses a probabilistic seismic demand model and Monte Carlo simulation to determine the ideal fluid viscous damper settings for reducing the responses of cable-stayed bridge. They examined the effectiveness of FVDs with parametric variations applied to fragility curves at the system and component levels. It was revealed from their research that some fluid viscous damper parameters can potentially negatively impact specific bridge components while also improving the system's overall effectiveness for cable-stayed bridges. The effect of a viscous-shear damper (VSD) on a long-span cable-supported bridge with different modes of vibration was examined by Chen et al. [167]. Shearing plates submerged in a viscous medium inside a casing box filled with liquid constitute a conventional VSD. The vibration energy is absorbed through shear deformation of the viscous medium. For the cable-supported Sutong Bridge, three VSDs were created in order to investigate the damping effects on long cables. These VSDs were tested in the lab using forced sinusoidal displacements at different frequencies and amplitudes.

Tell et al. [168] presented a vibration mitigation strategy for a simply-supported bridge by incorporating additional FVDs positioned between the deck and abutment, aimed at mitigating vibrations based on a random vibration mode. Their approach involved applying Monte Carlo simulations considering uncertainties related to the bridge-damper system, ensuring the robustness of the proposed vibration mitigation technique. This probabilistic evaluation emphasized the necessity of conducting a statistical analysis to predict

model uncertainty and acceleration limit for dynamic analyses. Yi et al. [169] organized research on a combined system utilizing both FVDs and negative stiffness devices to decrease deck displacement in bridges. Negative stiffness devices are used to enhance the fundamental vibration period of the bridge, thereby reducing spectrum acceleration. By adopting negative stiffness devices, designers have gained additional options for developing FVDs in cable-stayed bridges to meet design requirements for near-field earthquake excitations. To reduce transverse earthquake vibrations in reinforced concrete (RC) continuous girder bridges, Wang et al. [170] introduced a system employing multi-viscous inerter dampers (MVIDs), which are placed between girders and piers. They conducted a robustness analysis of optimally designed MVIDs under perturbation to the lateral stiffness of rubber bearings and damping ratios of viscous dampers. The findings of the analysis indicated that the mitigation efficiencies of MVIDs were more sensitive to changes in damping ratio than alterations in the lateral stiffness of rubber bearings.

He et al. [171] proposed a novel bionic viscoelastic limb-like-structure (VELLS) device and carried out investigation on its mechanical properties and energy dissipation ability. Based on their energy dissipation model, they suggested an improvised control strategy.

2.1.2. Real-life applications of passive control devices in bridges

The technique of seismic isolation, recognized as a leading and highly effective seismic mitigation method, finds extensive application in the construction of small-to-medium span bridges, particularly in countries located in active seismic zones such as the United States, Mexico, Japan, and China. Within the context of seismic isolation systems, the use of laminated rubber bearings plays a crucial role. These laminated rubber bearings not only provide substantial load-bearing capacity but also demonstrate remarkable versatility by accommodating various types of motion, including vertical and shear displacements. This adaptability serves to safeguard not only the structure itself but also the surrounding infrastructure, making laminated rubber bearings an integral component of seismic resilience strategies in these regions [172]. One of the most remarkable cases of application of laminated rubber bearings is Mianzhu Huilan Bridge, China (Figure 2). Lead rubber bearings (LRBs) are seismic isolation devices designed to absorb and dissipate energy during earthquakes, protecting structures from damaging vibrations. Their unique flexibility and load-bearing capacity make them a vital component in earthquake-resistant building and bridge design, and are predominantly used in countries like United States, New Zealand and Japan. Among the most successful installation of lead rubber bearing in humungous bridges across the world are in Santa Ana River Bridge, USA (see Figure 3) and Maungatapu Bridge, New Zealand [173].

High-Damping Rubber Bearings (HDRBs) are specialized seismic isolation designed with rubber compounds that possess high energy dissipation properties, enabling them to

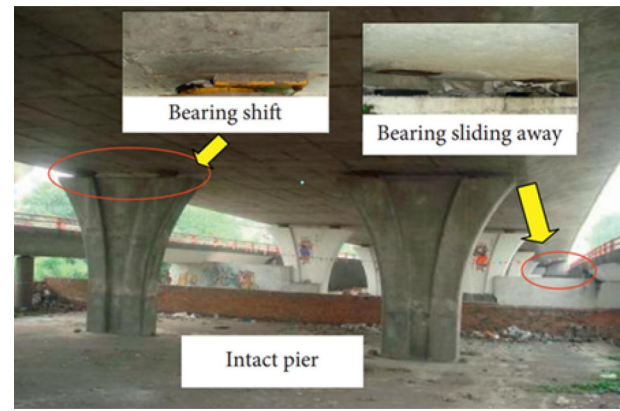


Figure 2. Laminated rubber bearing installed Mianzhu Huilan bridge, China [172].



Figure 3. Lead rubber bearing installed Santa Ana River bridge, USA [173].

absorb and dissipate seismic forces, thus protecting bridges and infrastructures from significant damage during seismic events. HDRBs are widely employed in earthquake-prone regions, notably in countries like Japan, Taiwan, and New Zealand, due to their exceptional ability to mitigate seismic forces and minimize structural damage [174]. Some notable bridges equipped with HDRBs are Imokawa Bridge in Japan and Sakaigawa Bridge in Japan (shown in Figure 4(a,b) respectively).

Researchers and engineers often assert that passive structural response control systems like Tuned Mass Dampers (TMDs) are regarded as straightforward and dependable. These systems employ mechanical components to dissipate a portion of the energy imparted to a structure, thereby diminishing structural response and potential damage. Certain passive systems have been subjected to thorough investigation and have established their efficacy in practical applications across various structures, such as the Millenium Bridge, England (Figure 5) and the Bannaguru Bridge, Japan [175]. Another remarkable passive vibration control device is the tune liquid damper (TLD) which is an inertia-based dissipation system that produces counteractive forces to balance the overall forces acting on the engineering structure of interest. In addition to being economical, these devices are easy to install and are being rigorously tested for robust application. Among the remarkable installation is the Higashi-Kobe Bridge, Japan (Figure 6) [176].



(a)



(b)

Figure 4. HDRB installed (a) Imokawa bridge, Japan; (b) Sakaigawa bridge, Japan. [174].



Figure 5. TMD installed Millenium bridge, England [175].



Figure 7. The Dong-Yun bridge, South Korea installed with FVD near each pier [177].



Figure 6. TLD installed Higashi-Kobe bridge, Japan [176].

Fluid viscous dampers excel in vibration mitigation by harnessing the resistance of viscous fluids to dissipate kinetic energy. On integration with structures, they counteract oscillations, enhancing safety and comfort. Their ability to provide both stiffness and damping makes them a versatile solution for enhancing structural stability and reducing dynamic loads. Among the most established cases of FVD installations are Dong-Yun Bridge (Figure 7), Chun-Su Bridge, Ok-Yeo Bridge, E-Po Bridge, Kang-Dong Bridge, etc. [177]. A viscous shear damper is an engineering device designed to mitigate structural vibrations and dissipate energy in buildings and bridges. It operates by utilizing the viscous resistance of a fluid, typically oil, to absorb and dissipate the kinetic energy generated during seismic events or



Figure 8. Viscous shear damper controlled Sutong bridge, China [167].

wind-induced oscillations. This technology plays a crucial role in enhancing the resilience and safety of structures, reducing the risk of damage or collapse during extreme conditions. Viscous shear dampers are a key component in modern structural engineering for improving structural stability and protecting against dynamic forces. Some of the eminent bridge applications of viscous shear damper are Sutong Bridge (Figure 8) [167], Fred Harten Bridge and the Stonecutters Bridge. Real-life applications of passive control devices in bridges are summarized in Table 1.

Table 1. Real-life applications of passive control devices in bridges.

Control system	Key feature	Real-life applications in bridges
Seismic isolation system		
Laminated Rubber Bearing	Comprises of rubber and metal plates in alternating layers	Vidyasagar Setu, Kolkata, India; Akashi Kaikyō Bridge (also known as the Pearl Bridge), Kobe, Japan
Lead Rubber Bearing (LRB)	Laminated rubber bearing is driven with a central lead core to enhance damping efficiency	San Francisco-Oakland Bay Bridge, California, USA; Vincent Thomas Bridge, California, USA
High-Damping Rubber Bearing (HDRB)	- Comprises of specially designed rubber with exceptional damping performance. - Sandwiched between steel plate layers. - Capable of absorbing vibrations through substantial displacement along the direction of external excitation	Hale Boggs Memorial Bridge (also known as Luling Bridge), Louisiana, USA; Awa Shirasagi Ohashi Bridge, Tokushima, Japan
Friction Pendulum System (FPS)	Configured for accommodating varying magnitudes of displacements, by altering the curvature and diameter of the bearing surface.	Gerald Desmond Bridge, California, USA
Shape Memory Alloy (SMA)	- SMAs have the potential to recover their original shape when strained beyond 6-8%. - This shape recovery is facilitated by a reversible phase transformation. - Exhibit outstanding potential for dynamic applications due to the hysteretic behavior linked to this transformation.	Mersey Gateway Bridge, Cheshire, United Kingdom
SMA- LRB	- NiTi SMA wire is employed to improve the re-centering ability of the lead rubber bearing (LRB). - In situations involving extremely high shear deformation, such as a 200% shear strain, the device may experience issues. - This malfunction occurs because the wires experience axial strain exceeding the super-elastic strain range of NiTi.	San Francisco-Oakland Bay Bridge East Span, San Francisco, USA; Yeongjong Grand Bridge, Incheon, South Korea
SMA- HDRB	- SMA wires serve as an additional component to improve HDRB performance. - They increase energy dissipation capacity and reduce residual deformation, especially under high shear strain amplitudes.	Jindo Bridge, South Jeolla, South Korea; Newer Span of Tacoma Narrows Bridge (also known as Galloping Gertie), Washington, USA.
Tuned damper		
Tuned-mass damper (TMD)	- Response control attachment comprising a mass, spring, and damper. - The damper's frequency is adjusted to match the structure's frequency. - This tuning allows the damper to resonate out of phase with the structural motion when the specific frequency is excited.	Aratsu Bridge, Fukuoka, Japan; Yokohama Bridge, Kanagawa, Japan; Millenium Bridge, London, England; Akashi Kaikyo Suspension Bridge, Kobe, Japan.
Pounding tuned mass damper (PTMD)	- PTMD enhances the classical Tuned Mass Damper (TMD) by incorporating a single or double-sided delimiter covered with viscoelastic material. - The delimiter serves a dual role: i. Restraining the motion of the tuned mass. ii. Increasing energy dissipation within the system.	Fuchu Lake Bridge, Kagawa, Japan
Viscous damper		
Fluid viscous damper (FVD)	- This system opposes motion through viscous friction. - The resultant force is directly proportional to velocity but acts in the opposite direction, thus decelerating the motion by absorbing the energy.	Vincent Thomas Bridge, California, USA;
Viscous-shear damper (VSD)	- It comprises of an annular seismic mass or inner member enclosed within a ring-shaped casing. - Separation between the seismic mass and casing is achieved through peripheral and lateral gaps or clearances filled with viscous fluid. - The importance of untuned dampers lies in their ability to lower the natural frequency of the system and simultaneously decrease vibration amplitudes.	Xia zhang Bridge, Xiamen, China; Shishou Changjiang Bridge, Hubei, China; E'dong Yangtze River Bridge, Hubei, China

2.2. Devices for semi-active vibration control system

In comparison to normal active control systems, these systems activate with less power. These systems use reactive components with changing stiffness and properties to produce opposing forces. They are also known as programmable passive systems. These systems can function during a strong earthquake since they only need a little amount of external energy, which may be provided by battery power. Because they do not introduce energy into the

structure, these semi-active control methods do not lead to structural instability. A schematic diagram of semi-active vibration control system of bridge structure is shown in Figure 9.

2.2.1. Bridge vibration control strategies using semi-active devices

Variable stiffness devices, controllable impact dampers, controllable friction devices, controllable fluid dampers, semi-

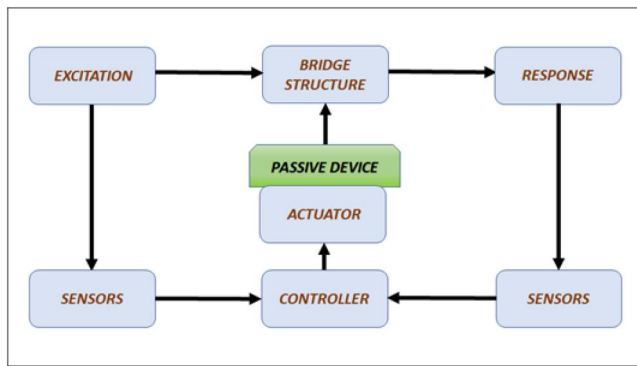


Figure 9. Schematic diagram of semi-active vibration control system of bridge structure.

active tuned liquid dampers, variable-orifice fluid dampers, and smart tuned mass dampers are all frequently used semi-active control devices. Symans and Constantinou [178] produced detailed studies on semi-active control systems that provided overviews of all the semi-active devices in use and emphasized the effectiveness of various devices in reducing structural vibration. The use of different semi-active dampers to mitigate the earthquake vibration of bridges was reported in the literature. As was previously noted, a sort of active control system known as a semi-active control system uses power just to modify the mechanical characteristics of the damper. Magnetorheological (MR) damper and semi-active tuned dampers are most widely used semi-active control devices in vibration control studies of bridges.

2.2.1.1. Magnetorheological (MR) damper. A mathematical model for the magnetorheological damper was introduced by Spencer et al. [179] based on a Bouc-Wen hysteresis model [180]. The model got over the many drawbacks of the prior mechanical variants. The model successfully illustrated a variety of hysteresis behavior and was numerically trackable. The arrangement of the dashpot and Bouc-Wen models in the model successfully depicted the non-linear force roll-off in the area where the velocity approached zero. An extra spring element helped the damper to account for the accumulator's rigidity. Jung et al. [181] presented the use of the MR damper in different structures. The study suggested new semi-active control strategies, including the control structure interaction. The execution of the mitigation strategy was evaluated for several earthquake ground motions. The study showed that semi-active control using an MR damper could efficiently control the seismic vibration of the cable-stayed Bridge. Ze-bing et al. [182] investigated the efficiency of an MR damper-based controller to mitigate the vibration of a benchmark cable-stayed bridge. The study also considered the multi-support excitation. The control efficiency of the semi-active controller was correlated with that of active control. The study reported that the seismic response was efficiently controlled by the semi-active controller, and it achieved the same level of performance as that of an active control system. Choi et al. [183] checked the efficiency of a passive system using an MR damper for vibration mitigation of a highway benchmark bridge problem. A permanent magnet and a coil-based electromagnetic

induction system were used by the controller, together with an MR damper system. The investigation validated the efficacy of the suggested control mechanism. A non-linear clipped feedback control approach based on optimum polynomial control (OPC) of a two-span bridge utilizing an MR damper was put forth by EI-Khoury et al. [184]. The study addressed the linearization of the non-linear behavior of different components. Using a genetic algorithm, the waiting matrices in the OPC objective function were chosen. The input current for the MR damper was controlled using a clipped stochastic optimum polynomial control technique. A unique real-time semi-active control method based on the MR damper's ability to measure damage to bridge members was suggested by Li et al. [185]. In order to regulate the vibration for a 3-span continuous reinforced concrete bridge modeled using non-linear fiber damage components, the efficacy of the controller was examined. Additionally, the effectiveness of passive control employing MR and viscous dampers with constant current was examined. The study's findings demonstrated that the suggested approach outperforms passive controls in terms of safeguarding the bearing and pier. In order to address the dynamic uncertainties of a benchmark cable-stayed bridge equipped with an MR damper, Yeganeh Fallah and Attari [186] introduced a reliable controller for bridges. The decoupled system's assessment output was regulated by the robust H controller, which was resistant to the effects of uncertainties, disturbances, and sensor noise. The findings showed that the suggested controller was successful in reducing the undesirable uncertainty impact on the response of the earthquake-exposed structure. There were also reports of other literatures demonstrating the MR damper's efficiency in lowering the seismic response of the bridge [187–191].

The performance of an MR damper can be managed only in terms of the command voltage or current that is applied to the damper. It is necessary to create a suitable inverse dynamic model in order to compute the voltage that must be given to the MR damper to produce the control forces. It is difficult to obtain the inverse dynamic model of the MR damper because of its nonlinear properties. Some inverse dynamic models for MR dampers have been offered as a remedy for this issue [192, 193]. Boada et al. [194] suggested a novel network-based dynamic model for the MR damper. A model was created to demonstrate the accuracy to estimate the voltage needed for providing the necessary control force. Their study showed that the new version of the inversion technique was effective for the inverse modeling of an MR damper. However, these models require precise parameters from the empirically determined Bouc-Wen model. An experimental study was carried out with MR damper for a lab based three span bridge to mitigate pounding phenomena. They proposed an innovative semi-active control strategy on the basis of sliding mode control with optimal polynomial approach. It was observed in their study that MR damper is efficient for seismic vibration mitigation between the adjacent bridge spans [195]. To mitigate the vibration of Lali cable stayed bridge, an analytical study was demonstrated with MR damper and two new algorithms of

semi-active Fuzzy Control [196]. Traditional MR dampers have a damping force that is extremely sensitive to changes in velocity. Furthermore, enormous sizes and complicated designs are common characteristics of traditional MR dampers. Liu et al. [197] proposed a unique double-rod bypass magnetorheological damper for civil engineering structures to overcome these constraints. A semi-active control method using a magnetorheological fluid damper (MRFD) was presented by Du et al. [198] to reduce the parametric vibration of very long stay cable. The findings of the analysis indicated that the parametric vibration of super-long stay cables in cable-stayed bridges may be reduced by using this semi-active control based MRFD. This semi-active control system can function similarly to the LQR-based optimum active control, according to the results of the analysis.

2.2.1.2. Semi-Active tuned damper. For the purpose of controlling structural vibration, a semi-active tuned mass damper (STMD) with adjustable damping or stiffness has been introduced. An adjustable damping device, such as a variable-orifice hydraulic device employing magnetorheological fluid, replaces a passive damping mechanism for this type of damper. Therefore, the damping or stiffness of such devices can be assembled with a low activation energy, excluding the requirement for massive energy storage that the active tuned mass damper (ATMD) needs. Thus, it offers a substitute for both active and passive TMD. Both the tuned liquid damper (TLD) and the tuned liquid column damper (TLCD), which possess a similar concept, provide indirect damping to the system and enhance structural performance. TLDs absorb structural energy through fluid viscosity and wave breaking, whereas TLCDs typically dissipate energy through the liquid's passage through an orifice that has inherent head loss characteristics. By continuously adjusting their stiffness and damping, controllable TMD's and TLD's aim to reduce the sensitivity of the aforementioned passive devices to frequency and damping variations of the primary structure. In practice, this is typically achieved by combining other semi-actively controlled mechanisms.

The advantages of implementing an STMD for the reduction of wind-induced vibration were examined by Hrovat et al. [199]. It was demonstrated that the STMD and ATMD have comparable efficiency subjected to wind loads under certain assumptions. Different types of TMDs were studied in order to reduce cable vibrations in cable-stayed bridges or, more often, aerodynamic instabilities (such as fluttering and buffeting) of decks, primarily in suspension bridges [200]. Two degrees of freedom TMD, which had the feature of tuning frequencies corresponding to the vertical and torsional modes of the bridge with damping regulated by a semi-active hydraulic damper, had shown a better performance in the semi-active approach compared to passive systems [201]. In order to mitigate the buffeting response of a long span cable-stayed bridge during construction, a semi-active TLCD with adaptive frequency tuning capacity (adjusted by regulating the air pressure at the two ends of a container) was investigated [202]. Also, an alternative semi-

active technique was presented to suppress wind-induced vibrations during construction. The electromechanical device utilized in the TMD control strategy was a pendulum connected to an alternator. This device's function was to transform the mechanical energy of the oscillating pendulum into electric energy, which was then dissipated *via* the joule effect through an external resistance [203]. A semi-active tuned mass damping system were studied for highway bridges in order to mitigate the earthquake-induced excitation [204]. To improve control efficacy, a unique magnetorheological elastomer variable stiffness TMD was placed in the steel bridge tower. It was observed in their study that the variable stiffness tuned mass damper with magnetorheological elastomer could improve passive TMD's performance by up to 40%. Furthermore, the softening behavior was observed to enhance the stiffness adjustable range of MRE layers by a factor of two [205]. To control vortex induced vibration for a MSC bridge, a magnetorheological TMD was proposed with control algorithm [206]. The efficiency of the semi-active tuned mass damper (STMD) to lessen resonant vibrations of rail bridges under high-speed trains was investigated by Nguyen et al. [207]. According to their study, the STMD not only performed better in resonance but also had a more extensive working range within the specified frequency range.

2.2.2. Real-life applications of semi-active control devices in bridges

Semi active dampers are the adaptive amalgamation of passive and active vibration control features. MR dampers are one of the advanced semi active shock absorbers used in major bridges across the globe. The vibration control philosophy of Magnetorheological (MR) damper contains a special fluid filled with tiny iron particles. When a magnetic field is applied to the fluid, typically through an electromagnet, the iron particles align along the magnetic lines of force. As the alignment of particles changes in response to varying road conditions or external inputs, the viscosity of the fluid also changes instantaneously leading to variation in resistive stiffness.

MR dampers excel in seismic control due to their ability to adapt rapidly to changing ground motions. Unlike traditional dampers, MR dampers can adjust their stiffness and damping levels in real-time, offering superior performance in mitigating structural vibrations and ensuring the safety and stability of bridges during seismic events. Among the most successful installation of MR dampers in humungous bridges across the world are Eiland Bridge, Netherlands (Figure 10(a)), Shandong Binzhou Yellow River Highway Bridge in China, Russky Bridge in Russia (Figure 10(b)), Dongting Lake Bridge in China, Alamillo Bridge in Spain, etc. [208, 209].

A finer variant of semi-active damper is the Magnetorheological Seismic Tuned Mass Damper (MR-STMD), which operates by integrating magnetorheological (MR) fluid technology into a tuned mass damper system. The key characteristic of the MR-STMD concept lies in the dynamic control of the MR damper, which mimics both



Figure 10. MR damper controlled (a) Eiland bridge, Netherlands [208]; (b) Russky bridge, Russia [209].



Figure 11. Volgograd Bridge, Russia, equipped with magnetorheological seismic tuned mass damper [210].

adjustable stiffness and friction forces in real-time. The adaptable stiffness force supplements or reduces the stiffness provided by passive springs, aligning the MR-STMD's frequency with the bridge's actual frequency. Simultaneously, the variable friction force contributes to frequency-dependent energy dissipation. The experimental findings demonstrate that the precise stiffness emulation technique greatly improves the effectiveness of the MR-STMD, especially when it operates with lower desired damping levels. This improvement is particularly notable when control force constraints start to have a notable impact on the system. During seismic events, the MR-STMD adjusts its properties dynamically to counteract structural vibrations, effectively reducing resonance and dissipating seismic energy [210]. This responsive and adaptive mechanism enhances its seismic mitigation capabilities, providing efficient and precise control over bridge sway, minimizing damage, and ensuring the safety of structures and its occupants during event. The first MR-STMD was installed in Volgograd Bridge, Russia (Figure 11). Real-life applications of semi-active control devices in bridges are summarized in Table 2.

2.3. Devices for active vibration control system

Active control systems are more sophisticated assemblies that require a significant source of power for operation. These control systems use sensors to measure the structural responses. Based on the feedback from the sensors, the control forces are computed according to a prescribed controller and are then applied to the structures through actuators.

These systems have the advantage over passive systems in adapting to varying load conditions. However, it also can destabilize the structure of the control forces that are not applied in the proper position and at the proper time. An overview of smart materials and their structural applications was presented by Tzou et al. [211]. Some of the recent studies suggested innovative active vibration control systems employing piezo electric patches for refined performances [212, 213]. A schematic diagram of active vibration control system of bridge structure is shown in Figure 12.

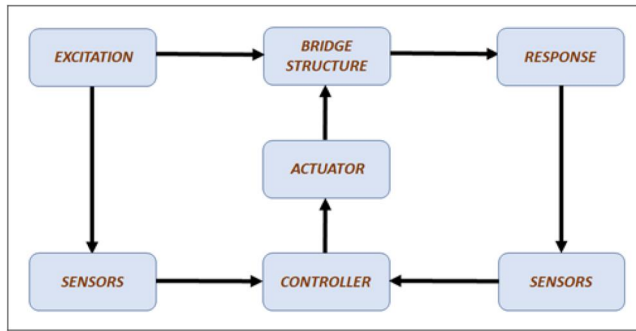
2.3.1. Bridge vibration control strategies using active devices

Some of the active control devices used in bridge structures are active tuned mass damper, active mass damper, and active tendon system. Since supplemental devices placed close to the cable anchorages, have been employed to lessen or eliminate damaging vibration of cable-stayed bridges. Recently, control solutions to actively reduce cable vibrations have been suggested. A method for vibration mitigation of cable-stayed bridges that combines wavelet feedback and linear mean squared model was proposed by Kim and Adeli [214].

Pereira et al. [215] presented an active vibration control strategy aimed at mitigating human-induced vibrations on pedestrian bridges through a multi-input multi-output vibration control approach. Most studies in vibration control primarily consider linear behavior of structures. Wang and Adeli [216] proposed for a novel self-constructing wavelet neural network approach using an adaptive nonlinear fuzzy

Table 2. Real-life applications of semi-active control devices in bridges.

Control system	Key feature	Real-life applications in bridges
Magnetorheological (MR) damper	- MR dampers share a construction resemblance with fluid viscous dampers, but it employs magnetically polarizable magnetorheological fluids within a carrier medium like mineral or silicone oil instead of hydraulic oil. - When subjected to an applied magnetic field, these particles change their form, causing the fluid to transition rapidly into a plastic or semisolid state within a few milliseconds due to the alignment of iron particles with the magnetic field.	Dongting Lake Bridge, Hunan, China; Binzhou Yellow River Bridge, Shandong, China; Alamillo bridge, Seville, Spain; Franjo Tudman bridge, Dubrovnik, Croatia; Russky bridge, Vladivostok, Russia
Semi-active tuned mass dampers	- Semi-active TMDs are essentially passive TMDs with adjustable features/mechanisms. - They automatically reconfigure their parameters to remain optimally tuned to their target frequencies. - Through continuous in-situ re-tuning, achieved by adjusting stiffness and/or damping, semi-active TMDs maintain robust performance even when the dynamics and natural frequencies of their target structures change.	Volgograd bridge, Volgograd, Russia
Semi-active tuned liquid dampers	- They utilize the movement of a liquid column within a U-shaped tube to counterbalance external forces affecting the structure. - Inherent damping is introduced into the oscillating liquid column via an orifice.	Walnut Creek bridge, Oklahoma, USA

**Figure 12.** Schematic diagram of active vibration control system of bridge structure.

sliding mode control technique for large structures. Mitchell et al. [217] introduced a wavelet filtered genetic neuro-fuzzy system incorporating discrete wavelet transforms (DWTs), evolutionary genetic algorithms [218, 219], neural networks, and fuzzy logic [220, 221] to regulate the dynamic behavior of highway bridges subjected to various seismic loads. Bossens and Preumont [222] utilized an active tendon system employing either hydraulic or piezoelectric actuators to regulate vibrations in two differently sized cable-stayed bridge models under wind-induced loads.

Pre-stressed tendons, known as active tendons, were used in cable-stayed bridges to act as cross braces. Actuators were installed to alter the tension in the cables and determine the amount of control force supplied to the structure [223]. Rodellar et al. [224] investigated the behavior of a cable-stayed pedestrian bridge (142.5-m long), managed by an active tendon control system. A controller based on Lyapunov subjected the simulated bridge to the 1952 Taft earthquake. Significant lowering of the bridge's response was possible due to the active tendon system. Inoue et al. [225] proposed an active mass damper (AMD) to mitigate tower VIV during the Osman Gazi bridge construction and in-service conditions. By offering more damping than what was required by the design specification, the AMD system was proven to be successful in reducing vibration in both situations.

2.3.2. Real-life applications of active control devices in bridges

In real life, active control is said to be better than passive devices in the circumstances as follows i) there are several vibration modes; ii) natural time period shifts, as is commonly seen during construction; and iii) there is limited installation area, thus compact devices are recommended. Activated tendon are mostly employed in cable-stayed bridges. The trigger controlled adaptive nature of active control systems makes it a superior choice over other dampers. The control strategy gained a huge acceptance in seismic prone island country of Japan. Some of the renowned examples of application of active damper system in bridges are Rainbow Bridge in Japan, Kurushima Bridge in Japan, Meikoh-Central Bridge in Japan and Tsurumi-Tsubasa Bridge in Japan. [226, 227].

Tendon control forces are produced by the hydraulic rams' movements, which lessen the vibration of the structure (see Figure 13(b)). The motion of hydraulic rams is controlled by the sensors and control algorithm. The Normandy Bridge in France (see Figure 13(a)), which has a span of 850 meters with control and stay cables installed, serves as the most significant example [228, 229]. The primary challenge with these systems is the nonlinear behavior of the cables, particularly when there are passing vehicles and concurrent wind loads. Real-life applications of active control devices in bridges are summarized in Table 3.

2.4. Devices for hybrid vibration control system

Hybrid vibration control systems are created by integrating either an active or semi-active system with a passive system. These systems utilize various control devices, combining both passive and active elements within a single structure. By merging these systems, it becomes possible to mitigate certain limitations and drawbacks inherent in each individual system when used in isolation. Consequently, this combined approach can enhance overall performance. Figure 14

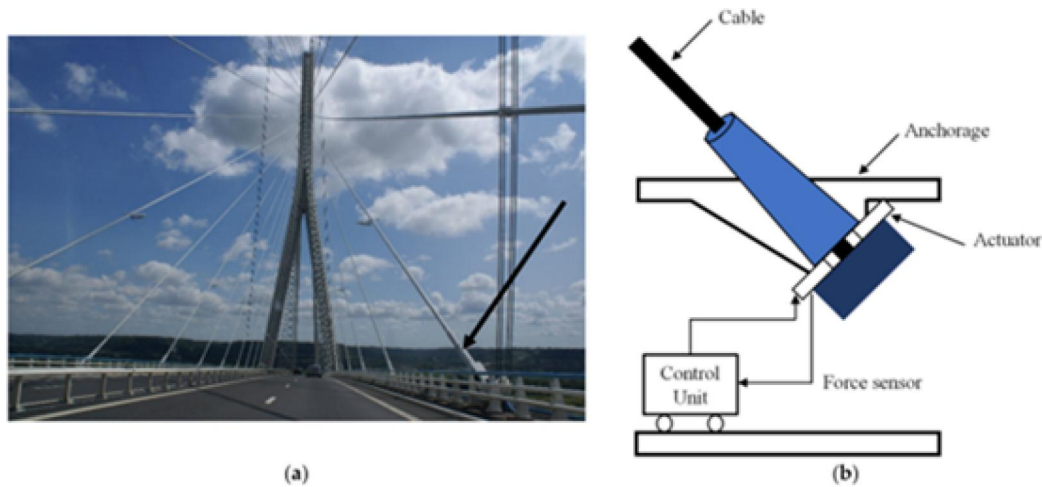


Figure 13. (a) Active tendon cable equipped Normandy bridge (b) schematic diagram of active tendon [228].

Table 3. Real-life applications of active control devices in bridges.

Control system	Key feature	Real-life applications in bridges
Active mass damper	• Involves a mass linked to the structure via an actuator. • Efficiency is constrained to the fundamental frequency. • Cost-effective for full-scale structures	Meiko-Chuo (Central) bridge, Nagoya, Japan; Akinada bridge, Hiroshima, Japan
Active tendon systems	• Utilizes pre-stressed tendons and electrohydraulic servomechanisms. • Applicable in both pulsed and continuous time modes. • Allows for the utilization of existing structural members, minimizing costs.	Still in research progress (experimental and analytical studies noted)
Active brace systems	• Utilizes existing structural braces, an actuator, and can be integrated into three common types of bracing systems.	Still in research progress (experimental and analytical studies noted)

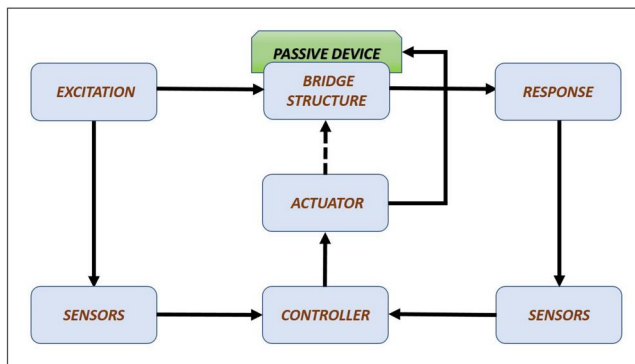


Figure 14. Schematic diagram of hybrid vibration control system of bridge structure.

illustrates a schematic diagram of a hybrid vibration control system implemented in a bridge structure.

2.4.1. Bridge vibration control strategies using hybrid devices

Bridges, a key component of transportation networks, is essential for connecting the network of traffic. As a result, when a bridge fails during an earthquake, there will be fatalities as well as secondary losses, including collisions, car breakdowns, and energy losses as a result of the extreme congestion brought on by network failure. For example, because of the variable motion of each higher spans, the Loma Prieta earthquake (US, 1989) and the Kobe earthquake (Japan, 1995) both severely damaged bridges and ultimately destroyed them [230]. Similarly, after the 2008 Wenchuan

earthquake, the Miaoziping and Gaoyuan Bridges collapsed due to failure of shear key [231]. Further, the Momence River Bridge's intermediate supports were affected by the 2010 Haiti earthquake [232]. As a result, it's critical to build bridges that can withstand environmental pressures like wind and earthquakes. Numerous researchers have developed various vibration-control methods during the past few decades to lessen undesirable bridge-structural vibration [233–236]. Some of the combinations that have been studied as hybrid vibration control devices are seismic isolation and active actuators, seismic isolation and tuned mass damper, seismic isolation and visco-elastic damper, etc.

In the last few years, seismic isolation devices have been installed on bridges in seismic zones. Large base-level movements that need to be decreased are a drawback of seismic isolation. Researchers have suggested hybrid seismic isolation systems as a solution to this problem. In recent decades, seismic isolation systems have undergone extensive research and practical implementation. Numerous studies have conducted comprehensive investigations into the behavior of structures employing base isolation [237–240]. A challenging issue in base isolation systems involves reducing the drift of the isolation system without increasing the complexity of the superstructures. Consequently, several studies have explored combining isolated systems with diverse supplementary damping mechanisms to address this challenge. A hybrid isolation system is one that combines fluid dampers with sliding bearings, or elastomeric bearings are applied [241]. Adaptive dampers can serve as replacements for

supplemental dampers in hybrid isolation systems, offering a potential solution to the hindrance faced by such systems. Therefore, isolators and MR dampers were integrated into much research. Nagarajaiah and Mate [242] presented a hybrid technology known as semi-active independently variable stiffness (SAIVS) that offered autonomously alterable stiffness in real-time. Later, an isolated structure's seismic response was controlled by the SAIVS system [243, 244]. Many other researchers investigated on adaptive isolation systems, and they addressed their effectiveness in bridge vibration mitigation [245, 246].

Many researchers have presented hybrid vibration-control techniques for cable-stayed bridges and reported on their performance [247–251]. The mitigation of vibration in other types of bridge structures was the focus of many studies [252–255]. Several investigations have been conducted to verify the effectiveness of innovative hybrid vibration-control systems [256, 257]. One experimental method that has gained wide acceptance in the testing field is the shaking table [258]. A vast range of seismic intensities were used to examine the reduction of dynamic responses of bridge with two different hybrid vibration control systems with a mixture of rubber bearings, sliding bearings, and actuators. They claimed that the hybrid technologies under consideration were particularly effective in reducing the bridges' dynamic response [259]. Iemura and Pradono [247] applied a seismic isolation technique, complemented by variable slip dampers, to substitute fixed-hinge connection between towers and deck of a cable-stayed bridge in Japan. In another research endeavor concentrating on seismic mitigation strategies for cable-stayed bridges, Jung et al. [260] assessed the successfulness of incorporating LRBs conjunction with linear passive viscous dampers. Their investigation revolved around evaluating the performance of this hybrid control system through numerical simulations, with the goal of enhancing the seismic resilience of cable-stayed bridges. Recently, an innovative hybrid vibration isolator that combines piezoelectric actuators and metal rubber with an adaptive active control technique was suggested by Jin et al. [261].

Additionally, Park et al. [249, 262] aimed to diminish the seismic response of a cable-stayed bridge by integrating an active controlled actuator with a passive base isolation system. They claimed that the suggested hybrid device have superior performance over the active and passive control device. A sliding isolator and an MR damper were coupled by Sahasrabudhe and Nagarajaiah [263] to lessen the response of a seismically agitated bridge. Through a shake-table test employing a laboratory-scaled bridge model and analytical investigations with multiple near-fault earthquake excitations, they established that MR dampers have a greater capacity to reduce bearing drift compared to passive systems with low or high damping. They even unveiled the semi-active hybrid system's isolator generated lower force than the highly damped passive systems. A study was conducted to investigate the use of hybrid and passive control for the protection of a reference cable-stayed bridge against seismic excitations. Based on numerical simulations, they parallelly

implemented viscously damped passive dampers with semi-active dampers and reported on the efficacy of the hybridized control mechanism [264].

For the purpose of controlling vibration in lightweight bridges, Gsell et al. [265] presented an adaptive TMD based on prestressable springs with piezo ceramic stack actuators. On the basis of their results from experiments, they summarized that the suggested tuned mass damper system could constantly adjust in a wide spectrum of frequencies. In order to protect a cable-stayed bridge against earthquakes, Soneji and Jangid [266] examined the efficacy of a semi-active hybrid control mechanism made up of a highly-damped rubber bearing and a magnetorheological damper. They claimed that the recommended hybrid device outperformed the passive hybrid device that used an isolator system with a passive viscous system. Soneji and Jangid [267] explored the viability of a hybrid control for a cable-stayed bridge that consists of a sliding and elastomeric isolation system with a FVD. They discovered that the inclusion of a FVD could minimize the base shear as well as the deformation of the isolator at the bridge tower. To mitigate the seismic response of a three-span straight bridge, Kori and Jangid [268] integrated magnetorheological dampers with laminated rubber bearings. They claimed that the proposed semi-active vibration control system had successfully reduced the seismic response. Saha and Jangid [269] aimed to mitigate the earthquake response of a standard cable-stayed bridge using a passive hybrid system. They explored various hybrid configurations, by combining highly damped rubber bearings, friction pendulum systems, robust friction seismic isolators, lead rubber bearings, and both linear and nonlinear viscous fluid dampers. Their parametric analysis revealed that the combination of a robust friction seismic isolator with a passive nonlinear viscous fluid damper exhibited superior performance compared to other hybrid configurations. Furthermore, Monzon et al. [270] conducted an investigation aimed at mitigating the dynamic response of a curved bridge, considering scenarios where the bridge was partially and fully isolated. Through experimental research, they found that both full and hybrid isolation systems effectively protected the bridge columns. In order to minimize the seismic response of a bridge with a horizontal curve, Kataria and Jangid [271, 272] focused on the efficacy of a hybrid device. They discovered that the hybrid device with MR dampers and lead rubber bearings resulted in a reduction in the curved bridge's seismic response. Furthermore, some researches were organized for bridges with hybrid vibration control systems that integrated high damping rubber bearings, with tuned-mass dampers, and additional viscous dampers. This integrated approach aimed to mitigate unwanted vibrations in the taut cables of cable-stayed bridges [273, 274].

In order to control bridges' seismic reaction, Krishnamoorthy [275] looked into the effectiveness of a system that combined viscous fluid dampers with a variable-radius friction pendulum. It was concluded that adding additional damping could reduce bridge vibrations. The proposed hybrid system was found to be superior to viscous

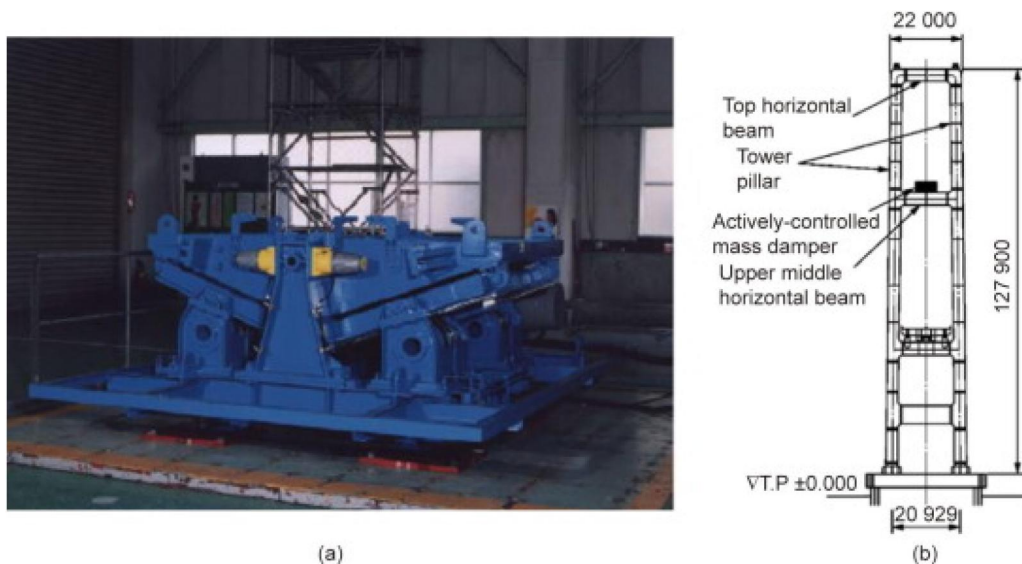


Figure 15. Vibration control system installed on the Hakucho bridge tower: (a) control device (pendulum type), and (b) installation location on the Hakucho bridge tower (unit in mm) [283].

fluid dampers. Domaneschi et al. [276] focused on implementation of various vibration control systems and a standard friction pendulum system. They looked at semi-active and passive control strategies for decreasing the undesired vibrations in typical long and medium span cable-stayed bridges. They looked at the finite element models of cable-supported and suspension bridges subjected to wind and earthquake excitations. The efficacy of the suggested control measures in minimizing the effects of high loading was documented. An analytical investigation was carried out to check the efficiency of Tuned-Mass dampers in minimizing the dynamic response of base isolated bridges, considering the interaction between the structure and the soil. They discovered that the soil around the bridge pier had a substantial impact on bearing displacement and that TMDs were very capable of lowering the dynamic responses of isolated bridges [277]. For the purpose of reducing the seismic response of a two-span bridge, Heo et al. [278, 279] coupled magnetorheological damper and the rubber bearing. The two-span bridge's seismic response was successfully decreased by the proposed hybrid system, according to experimental and computational simulations. In a recent study, researchers undertook an investigation to evaluate the efficacy of different supplementary dampers in mitigating significant displacements of isolated bridges during earthquakes. Their findings revealed that incorporating viscous fluid dampers alongside a FPS or LRBs for seismic isolation led to enhanced performance of bridges. This setup led to minimized bearing displacements, and decreased pier base shear levels during near-fault seismic excitations [280]. The damping properties of the hybrid application of a viscous damper (VD) and a tuned mass damper (TMD) on a taut cable of a cable-stayed bridge are studied by Yang et al. [281]. The study came to the conclusion that the suggested damper optimization approach for cable multimode vibration reduction may effectively achieve the optimal modal damping ratios in order to meet the damping requirement.

2.4.2. Real-life applications of hybrid control devices in bridges

The Hakucho Bridge is among the bridges that equipped with hybrid mechanism. Near the top of the tower, as seen in Figure 15(b), was a control device resembling the pendulum that is depicted in Figure 15(a). This device, which is commonly recognized as a "hybrid control device," combines the active control force produced by an electric motor-driven rack and pinion with the passive control effects of the pendulum motion [282, 283].

Real-life applications of hybrid control devices in bridges are summarized in Table 4. Among the notable examples of application of hybrid dampers in bridges are Tsurumi Tsubasa Bridge, Japan, Hakucho Bridge, Japan and Nakajima Bridge, Japan [226].

3. Comparative study of different bridge vibration control devices

According to the types of control schemes, the control device may be grouped into the following categories: passive, semi-active, active and the devices for hybrid vibration control systems, as discussed in the previous section (see Sections 2.1, 2.2, 2.3, 2.4). Bridge passive vibration control systems mostly depend on the inherent characteristics of control devices and the structure's capacity to absorb or release excess energy during a dynamic event (seismic or wind). They entail applying control current either continuously or inactively. According to the controller design, system characteristics, and performance function, active vibration control systems for bridges dynamically modify input signals to control devices. Due to the drawbacks of the passive and active vibration control systems mentioned above, intelligent devices such as hybrid and semi-active devices have been developed.

Semi-active vibration control systems apply a smaller amount of power to activate than active control systems. Reactive parts with varying stiffness and characteristics are

Table 4. Real-life applications of hybrid control devices in bridges.

Control system	Key feature	Real-life applications in bridges
Hybrid seismic isolation system	• Incorporates a seismic isolation system with active tendon control. • Combines a seismic isolation system with MR (Magnetorheological) fluid dampers.	Tsurumi Tsubasa Bridge, Yokohama, Japan; Sutong Bridge, Jiangsu, China
Hybrid mass damper (HMD)	• Combines TMD with an active control actuator • Cost-effective and requires low activation forces • Suitable for applications with limited space	Rainbow bridge, Minato, Japan; Nakajima bridge, Koshigaya city, Japan; Akashi Kaikyo, Kobe, Japan
Hybrid damper actuator bracing control	• Utilizes hydraulic actuators along with passive devices. • Mounted using K-braces on the structure. • Compatible with various passive devices in the system. • Option to integrate damper and actuator or keep them separate. • Cost-effective compared to a base-isolated/actuator system.	Still in research progress (experimental and analytical studies noted)

Table 5. Comparison of bridge vibration control systems and devices.

Category	Passive vibration control systems and devices	Active vibration control systems and devices	Semi-active control systems and devices	Hybrid control systems and devices
Mechanism	Energy dissipation or vibration mitigation through inherent mechanical properties or deformation	Sensors and actuators actively counter vibrations using external power	Adjusts properties like fluid viscosity or damping parameters in real-time	Combines passive and active/semi-active components for integrated performance
Advantages	Simple, cost-effective, no external energy required	Highly effective, adaptable to varying and unpredictable loads	Moderate energy use, adaptable, reliable in failure	Optimized performance, adaptable to dynamic conditions
Adaptability	Passive systems are fixed and best for predictable scenarios	Active and semi-active systems are suitable for variable probabilistic loads		Hybrid systems provide a balance between adaptability and simplicity
Energy Requirements	Passive systems require no external power.	Active systems demand significant energy input	Semi-active and hybrid systems operate with moderate energy needs	
Cost and Complexity	Passive dampers are cost-effective and simple	Active, semi-active and hybrid systems are costly and require advanced maintenance		
Effectiveness	Effectiveness depends on the deterministic design parameter and mode of dissipation	Active systems are most precise but energy-dependent	Semi-active systems balance precision and efficiency	Hybrid systems excel in both static and dynamic conditions
Limitations	Fixed damping characteristics; may not adapt to varying conditions	Expensive, requires power, complex systems, prone to failure	Less effective than fully active systems for extreme vibrations	High cost, complex design, maintenance-intensive
Superiorities Over Others	Reliable in low-maintenance scenarios, cost-efficient for consistent loads	Adaptable and precise for high-performance needs	Combines adaptability with energy efficiency	Balances adaptability, robustness, and energy efficiency

used in semi-active vibration control systems to generate opposing forces. They can alter structural parameters like the damping coefficient or spring stiffness, but they cannot add or subtract energy from the bridge. When controlling higher structural vibration modes, hybrid vibration-control systems are the most effective damping mechanisms. In the case of passive vibration control, devices with greater damping cannot satisfy vibration-control standards.

Bridge performance was examined using a hybrid vibration control system that included passive and semi-active [264, 266–272], passive and active [249, 259, 262] or semi-active and active vibration control systems. According to Park et al. [249], it merges the dependability and reliability of passive vibration control systems with the flexibility and adaptability of active vibration control techniques. Both the passive and active vibration control forces, which are greater control forces, can be used in hybrid control, but the active control's capacity cannot be used in semi-active control. This is an advantage of hybrid vibration control systems over semi-active vibration control systems. Due to the involvement of the active vibration control technique, it requires a significant quantity of external power, and

associated expenses are seen as a significant concern. It has been demonstrated that these advanced devices, such as hybrid and semi-active vibration control devices, function better than passive and active vibration control devices. Controlling the higher modal response is crucial for flexible constructions due to the comfort level. Compared to its passive equivalents, hybrid vibration-control systems are less commonly used, which may be because of the additional costs, maintenance requirements, and design challenges. Table 5 presents a categorical comparison of bridge vibration control systems and devices.

4. Summary and scope of further research

The primary objective of the present study is to present a comprehensive review of advancements in different vibration control devices, spanning from initial approaches to recent upgradations. It highlights the latest innovations in vibration control devices over the past two decades, highlighting the developments. Multiple literatures pertaining to vibration control devices for bridges have thoroughly been studied and an overview of the same has been provided in the

present study. Additionally, a critical analysis of the various bridge vibration control devices has been depicted along with their real-life applications.

From the review of literature, it is observed that the bridge vibration control devices are categorized into four main sections: passive, semi-active, active and the devices for hybrid vibration control systems. All the studies in the reviewed literature are further placed into specific sub-categories, and their achievements and enhancements in due course of time are discussed. The key features of each research, namely, the novel contribution, as well as the advantages and drawbacks are thoroughly studied in the present study.

In order to find the most efficient real-life application for bridge vibration control devices, understanding the dynamic behavior of bridges is of utmost importance. While passive control systems are extensively utilized currently, they are often tailored to specific dynamic movements and are constrained by limited efficacy within certain bandwidths. In recent years, active controls have gotten a lot of attention, with a variety of control approaches being developed to improve their performance under dynamic excitations. While passive dampers typically come with a higher initial cost compared to most active dampers, their long-term cost-effectiveness is evident due to their lack of reliance on additional energy sources for operation.

Presently, more advanced semi-active and hybrid vibration control devices for bridges have garnered significant attention from researchers owing to their dual functionality. Since semi-active and hybrid control devices provides more practical implementation strategies and integrate the benefits of active, and passive control devices, the progressing overall trend is toward hybrid and semi-active devices, over the traditional passive or active control devices. Researchers are becoming more interested in hybrid and semi-active control devices because of their improved real-world applicability and cause-specific usefulness. Based on the present review of literature, the following may recommended as scope of future research endeavors:

- Since TMDs are only tuned to the dominant frequency, they are vulnerable to inaccuracies during severe earthquakes. Nonetheless, effort should be made to tune the dampers to frequencies of multiple modes of bridges. Although the theoretical distribution of the multiple TMDI has been shown to be workable and efficient, the real-world difficulties in putting it into practice have not yet been thoroughly investigated. Nonetheless, further investigations and experimental verifications will validate its usefulness and practicality in real-world situations.
- Since bridges are long structures, research efforts should be directed toward developing control strategies that account for the spatial variation of ground excitations so that more accurate understanding and prediction of the behavior and performance of the vibration control devices can be made. Also, there is a need for research to take into consideration the soil-structure interaction

(SSI) in order to predict the performance of the structure-controller system more accurately.

- While assessing the most appropriate design of dampers and their placement under random excitations, it is essential to account for uncertainty at all levels. In order to take into consideration the uncertainties involved, the design of the control devices and the evaluation of their performance may be made based on stochastic approach instead of the conventional deterministic approach for better performance of these devices.
- To optimize the design parameters of vibration control devices, more appropriate optimization methodologies need to be used. Robustness of the optimized devices may be improved by further refinement of the optimized parameter values using the advanced techniques based on artificial intelligence-machine learning (AI-ML).
- Further research may be carried for the development of new vibration control devices for bridges based on improvised and innovative mechanical configuration of constituent structural components of the devices. New devices using state-of-the art advanced materials with required material characteristics may also be developed.

Authors' contributions

Tathagata Banerjee: Conceptualization, Literature review, Writing - Original Draft. Diptarka Ghosh: Conceptualization, Literature review, Writing - Original Draft. Diptesh Das: Supervision, Writing - Review & Editing.

Disclosure statement

No potential conflict of interest was reported by the authors.

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